

Living Person Presence Sensing Based on Non-Coherent MIMO Radar

Peichao Wang^{†,‡}, *Member, IEEE*, and Qian He^{‡,*}, *Senior Member, IEEE*

[†]College of Computer and Information Sciences, Fujian Agriculture and Forestry University, Fuzhou 350002, China

[‡]Department of Electronic Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, Sichuan 611731 China

Abstract—This paper investigates the problem of living person presence sensing using multiple-input multiple-output (MIMO) radar with possibly widely separated antennas, where the target in the monitored area may be either living or non-living. Exploiting the oscillatory nature of respiration, we develop a generalized received signal model with unknown respiration parameters and distinct reflection coefficients across different paths and hypotheses, unlike our previous studies that assumed an identical reflection coefficient. Conventional stationary target and coherent detectors are ineffective under these conditions due to time-varying delays, unknown respiration, and multi-dimensional reflection coefficients. In this case, we propose two non-coherent MIMO radar based living person detectors tailored for practical scenarios with known and unknown human reflection. Theoretical analysis and numerical experiments confirm the superior performance and robustness of the proposed detectors in living presence sensing.

Index Terms—MIMO radar, non-coherent processing, vital sign detection.

I. INTRODUCTION

In recent years, the increasing incidence of natural disasters such as earthquakes and mudslides has posed significant challenges to global emergency response systems. In such scenarios, a critical requirement is the rapid identification of survivors, which directly relates to the problem of distinguishing living persons from non-living objects. Unlike presence detection that solely verifies the existence of a target, living person detection exploits measurable vital signs to distinguish living persons from inanimate objects. To address this, various non-contact monitoring techniques have been studied, including laser-based sensing [1], vision-based approaches [2], infrared thermography [3], and electromagnetic methods [4]. Among these, radar sensing has received considerable attention due to its ability to capture weak physiological movements even through obstructions, making it especially suitable for disaster rescue operations [5]. Early studies employed single-input single-output (SISO) radar systems for respiration monitoring [6]–[8]. More recently, multiple-input multiple-output (MIMO) radar has emerged as a promising solution by exploiting spatial diversity [9], [10], enabling more accurate estimation of person vital signs [11], [12]. Motivated by these advances, this paper investigates the application of MIMO radar to the task of living person presence sensing.

The work of P. Wang was supported Foundation for Cultivated Young Talents of Fujian Province, China (No. 2026350245). The work of Q. He was supported by the Natural Science Foundation of Sichuan Province under Grant No. 2026NSFSC0417. (*Corresponding author: Qian He.)

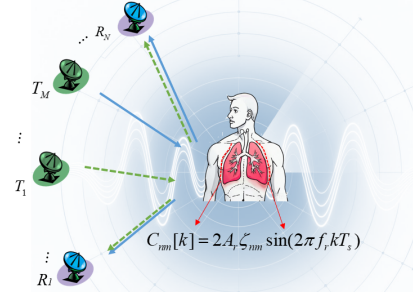


Fig. 1. Living person detection using MIMO radar.

In practical scenarios, key parameters associated with vital signs, such as respiration frequency, chest displacement, and reflection coefficients, are typically unknown, making living person detection a composite hypothesis testing problem. Earlier studies have considered signal-strength-based approaches [13], but such methods typically require calibration or prior knowledge, which limits their practical applicability. To address uncertainties in variance, the uniformly most powerful test (UMPT) was introduced in the context of voice activity detection [14]. However, UMPT is restricted to one-sided hypothesis testing [15] and is not generally available. As a more versatile alternative, the generalized likelihood ratio test (GLRT) has been widely adopted for composite hypothesis testing by substituting unknown parameters with their maximum likelihood estimates [16]. Unlike our previous work that assumed an identical reflection coefficient [9], this paper considers distinct reflection coefficients across different paths and hypotheses, which invalidates the assumption of coherent processing like matched filtering. For the first time, we derive two non-coherent MIMO radar detectors for living person detection, applicable to known and unknown human reflection scenarios, addressing the challenging of varying path reflections, time-varying delays, and unknown respiration, and further evaluate their detection performance.

II. SIGNAL MODEL

Consider a MIMO radar system with M transmitters and N receivers for living person detection, with tasks such as searching for a specific person or rescuing an unknown living person, as illustrated in Fig. 1. The low-pass equivalent signal emitted by the m th ($m = 1, \dots, M$) transmitter is expressed as $s_m[k] = \sqrt{E_m} s_m(kT_s)$, where E_m denotes the transmitted energy, k ($k = 1, \dots, K$) is the sample index and T_s is

the sampling interval. Each waveform is normalized such that $\sum_{k=1}^K |s_m(kT_s)|^2 T_s = 1$. Then overall $NK \times 1$ received signal vector is constructed as

$$\mathbf{y} = (\mathbf{y}_1^\dagger, \dots, \mathbf{y}_N^\dagger)^\dagger. \quad (1)$$

The received signal vector at the n th receiver is defined as

$$\mathbf{y}_n = (y_n[1], \dots, y_n[K])^\dagger, \quad (2)$$

where " $\dagger$ " denotes transpose, the signal received by the n th receiver ($n = 1, \dots, N$) at time kT_s ($k = 1, \dots, K$) consists of the echo from the target superimposed with clutter and noise, which can be modeled as

$$y_n[k] = \sum_{m=1}^M \frac{\sqrt{E_m}}{d_{nm}^2[k]} \xi_{nm} s_m(kT_s - \tau_{nm}[k]) \times e^{-j2\pi f_c \tau_{nm}[k]} + w_n[k], \quad (3)$$

where f_c is carrier frequency. The reflection coefficient associated with the nm th path is $\xi_{nm} \in \{\xi_{0nm}, \xi_{1nm}\}$, where ξ_{1nm} corresponds to a living person target, while ξ_{0nm} represents a non-living (dead or fake) target. While raw clutter may be correlated, it is effectively suppressed through preprocessing. Thus, the clutter-plus-noise $w_n[k]$, assumed to be white, is modeled as a complex Gaussian random variable, i.e., $w_n \sim \mathcal{CN}(0, \sigma_w^2)$. The σ_w^2 is known, which can be estimated using samples from target-free range bins. The $d_{nm}[k]$ denote the total propagation distance for the nm th path reflected by the target at time kT_s .

If a living person is present within the surveillance region, the periodic chest motion induced by respiration introduces a time-varying delay $\tau_{nm}[k]$ corresponding to the nm th path

$$\tau_{nm}[k] = \frac{d_{nm}[k]}{c} = \frac{d_{0nm}}{c} + \frac{C_{nm}[k]}{c}, \quad (4)$$

where the total propagation distance of the transmitted signal along the nm th path at time kT_s is given by $d_{nm}[k] = d_{0nm} + C_{nm}[k]$, with d_{nm} representing the nominal distance. The time-varying component $C_{nm}[k]$, induced by the chest motion due to respiration [17], is modeled as

$$C_{nm}[k] = 2A_r \zeta_{nm} \sin(2\pi f_r kT_s) \quad (5)$$

with $A_r > 0$ representing the peak chest displacement, ζ_{nm} the relative amplitude attenuation factor of the nm th path ($0 \leq \zeta_{nm} \leq 1$), such that $A_r \zeta_{nm}$ characterizes the effective respiratory amplitude along that path, and f_r denoting the respiration frequency. Since the nominal distance of each radar transceiver is much larger than the chest displacement, i.e., $d_{0nm} \gg 2A_r \zeta_{nm} \sin(2\pi f_r kT_s)$, the denominator of the weighted terms in (3) can be approximated as $d_{nm}^2[k] \approx d_{0nm}^2$, which is reasonable for subsequent analysis.

Then, the received signal (3) in the case where a living person is present can be expressed as

$$y_n[k] = \sum_{m=1}^M \frac{\sqrt{E_m}}{d_{0nm}^2} \xi_{1nm} s_m(kT_s - \tau_{1nm}[k]) \times e^{-j2\pi f_c \tau_{1nm}[k]} + w_n[k] \\ = \sum_{m=1}^M \xi_{1nm} s_{1nm}[k] + w_n[k]. \quad (6)$$

In the case of a non-living target without respiration, such as a corpse or a dummy, the propagation distance $d_{nm}[k] = d_{0nm}$

and the corresponding time delay along each path remain fixed over time

$$\tau_{nm}[k] = \tau_{0nm} = \frac{d_{0nm}}{c}, \quad (7)$$

and the reflection coefficient ξ_{0nm} associated with the nm th path of the non-living target is assumed to be deterministic but unknown. Then, the received signal (3) in the case where a non-living target is present can be expressed as

$$y_n[k] = \sum_{m=1}^M \frac{\sqrt{E_m}}{d_{0nm}^2} \xi_{0nm} s_m(kT_s - \tau_{0nm}) \times e^{-j2\pi f_c \tau_{0nm}} + w_n[k], \\ = \sum_{m=1}^M \xi_{0nm} s_{0nm}[k] + w_n[k]. \quad (8)$$

III. NON-COHERENT LIVING PERSON DETECTION WITH UNKNOWN RESPIRATION

The objective of living person detection is to distinguish a living person from a non-living target. Building upon the signal formulations in (1)–(8), the detection problem can be formulated as a binary composite hypothesis testing problem

$$\begin{cases} H_0 : \mathbf{y} = \mathbf{S}_0 \boldsymbol{\xi}_0 + \mathbf{w} \\ H_1 : \mathbf{y} = \mathbf{S}_1 \boldsymbol{\xi}_1 + \mathbf{w}, \end{cases} \quad (9)$$

where H_0 corresponds to the case of a non-living target, while H_1 denotes the presence of a living person. The $\mathbf{w} = (\mathbf{w}_1^\dagger, \dots, \mathbf{w}_N^\dagger)^\dagger$ is $NK \times 1$ clutter-plus-noise vector with $\mathbf{w}_n = (w_n[1], \dots, w_n[K])^\dagger$. For $i = 0$ or 1 ,

$$\boldsymbol{\xi}_i = (\xi_{i11}, \xi_{i12}, \dots, \xi_{iNM})^\dagger, \quad (10)$$

$$\mathbf{S}_i = \text{blk diag} \{ \mathbf{S}_{i1}, \dots, \mathbf{S}_{iN} \}, \quad (11)$$

which is a $KN \times MN$ delayed signal matrix, $\text{blk diag}\{\cdot\}$ is block diagonal matrix operator,

$$\mathbf{S}_{in} = \begin{bmatrix} s_{in1}[1] & \cdots & s_{inM}[1] \\ \vdots & \cdots & \vdots \\ s_{in1}[K] & \cdots & s_{inM}[K] \end{bmatrix}, \quad (12)$$

where $s_{inm}[k]$ is defined in (6) and (8). It is worth noting that when $\xi_{0nm} = 0$ for all $m = 1, \dots, M$ and $n = 1, \dots, N$, the detection task in (9) simplifies to presence/absence detection.

According to (9), the observation vector $\mathbf{y}$ under H_i ($i = 0$ or 1) follows the complex Gaussian distribution,

$$\mathbf{y} | H_i \sim \mathcal{CN}(\mathbf{S}_i \boldsymbol{\xi}_i, \mathbf{Q}), \quad (13)$$

where $\mathbf{Q} = \sigma_w^2 \mathbf{I}$. Hence, the likelihood functions under H_0 and H_1 can be written as

$$p(\mathbf{y} | H_0; \boldsymbol{\xi}_0) = \frac{1}{\pi^{KN} \det(\mathbf{Q})} \exp \left[-(\mathbf{y} - \mathbf{S}_0 \boldsymbol{\xi}_0)^H \mathbf{Q}^{-1} \right. \\ \left. \times (\mathbf{y} - \mathbf{S}_0 \boldsymbol{\xi}_0) \right] \quad (14)$$

and

$$p(\mathbf{y} | H_1; \boldsymbol{\xi}_1, A_r, f_r) = \frac{1}{\pi^{KN} \det(\mathbf{Q})} \exp \left[-(\mathbf{y} - \mathbf{S}_1 \boldsymbol{\xi}_1)^H \mathbf{Q}^{-1} \right. \\ \left. \times (\mathbf{y} - \mathbf{S}_1 \boldsymbol{\xi}_1) \right], \quad (15)$$

where $(\cdot)^H$ denotes complex conjugate transpose.

Since the respiratory signal and the reflection coefficient vector ξ_0 for the non-living target are generally unknown, and the human reflections ξ_1 may be known or unknown in different scenarios, the detector cannot exploit the exact phase information of the reflection coefficients. As a result, the proposed detector operates in a non-coherent manner. Next, we discuss the non-coherent detectors for these two scenarios.

A. Known Human Reflection Coefficients

For the case of searching for a specific person, where the human reflection coefficient vector ξ_1 is known while the non-living target coefficient vector ξ_0 , A_r , and f_r remain unknown, from (14) and (15), the corresponding LLR function is

$$L(\mathbf{y}; \xi_0, A_r, f_r) = \ln p(\mathbf{y}|H_1; A_r, f_r) - \ln p(\mathbf{y}|H_0; \xi_0) \\ = \left(\mathbf{y}^H \mathbf{Q}^{-1} \boldsymbol{\mu}_1 + \boldsymbol{\mu}_1^H (\mathbf{Q}^{-1})^H \mathbf{y} - \boldsymbol{\mu}_1^H \mathbf{Q}^{-1} \boldsymbol{\mu}_1 \right. \\ \left. - \mathbf{y}^H \mathbf{Q}^{-1} \mathbf{S}_0 \xi_0 - \xi_0^H \mathbf{S}_0^H (\mathbf{Q}^{-1})^H \mathbf{y} + \xi_0^H \mathbf{S}_0^H \mathbf{Q}^{-1} \mathbf{S}_0 \xi_0 \right), \quad (16)$$

where $\boldsymbol{\mu}_1 \triangleq \mathbf{S}_1 \xi_1$ is a constant and $\ln p(\mathbf{y}|H_1; A_r, f_r)$ is obtained under the case that ξ_1 in (15) is known.

Lemma 1. For a MIMO radar system with M transmitters and N receivers, consider the living person detection problem formulated in (9). In the case of a potentially present specific living person, where the reflection coefficient vector ξ_1 is assumed to be known whereas the peak chest displacement A_r , the respiration frequency f_r , and the non-living target coefficient ξ_0 are unknown, the corresponding non-coherent GLRT-based living person detector is given by

$$T_{\text{SGLRT}}(\mathbf{y}) \\ = \max_{A_r, f_r} \left\{ 2 \Re\{\mathbf{y}^H \boldsymbol{\mu}_1\} - \|\boldsymbol{\mu}_1\|_2^2 \right\} - \frac{|\mathbf{S}_0^H \mathbf{y}|^2}{\mathbf{S}_0^H \mathbf{S}_0} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma_{\text{SGLRT}}, \quad (17)$$

where $T_{\text{SGLRT}}(\mathbf{y})$ the test statistic, $\mathbf{S}_1$ and $\mathbf{S}_0$ the delayed signal matrices given in (11), and γ_{SGLRT} the decision threshold determined by the NP criterion.

Proof. The proof follows from using (16) and is omitted. $\square$

B. Unknown Human Reflection Coefficients

In the case of rescuing an unknown living person, where the parameters A_r , f_r , ξ_1 , and ξ_0 are unknown, the LLR function by referring to (14) and (15) is given by

$$L(\mathbf{y}; \xi_0, \xi_1, A_r, f_r) = \ln p(\mathbf{y}|H_1; \xi_1, A_r, f_r) - \ln p(\mathbf{y}|H_0; \xi_0) \\ = \left(\mathbf{y}^H \mathbf{Q}^{-1} \mathbf{S}_1 \xi_1 + \xi_1^H \mathbf{S}_1^H (\mathbf{Q}^{-1})^H \mathbf{y} - \xi_1^H \mathbf{S}_1^H \mathbf{Q}^{-1} \mathbf{S}_1 \xi_1 \right. \\ \left. - \mathbf{y}^H \mathbf{Q}^{-1} \mathbf{S}_0 \xi_0 - \xi_0^H \mathbf{S}_0^H (\mathbf{Q}^{-1})^H \mathbf{y} + \xi_0^H \mathbf{S}_0^H \mathbf{Q}^{-1} \mathbf{S}_0 \xi_0 \right). \quad (18)$$

Lemma 2. For a MIMO radar system with M transmitters and N receivers, consider a living person detection problem described in (9). In the case of a potentially present unknown living person with unknown but deterministic peak chest displacement A_r , respiration frequency f_r , human reflection coefficients ξ_1 , and non-living target coefficients ξ_0 , the non-coherent GLRT-based living person detector is

$$T_{\text{GLRT}}(\mathbf{y}) \\ = \max_{A_r, f_r} \left\{ \frac{|\mathbf{S}_1^H \mathbf{y}|^2}{\mathbf{S}_1^H \mathbf{S}_1} \right\} - \frac{|\mathbf{S}_0^H \mathbf{y}|^2}{\mathbf{S}_0^H \mathbf{S}_0} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma_{\text{GLRT}}, \quad (19)$$

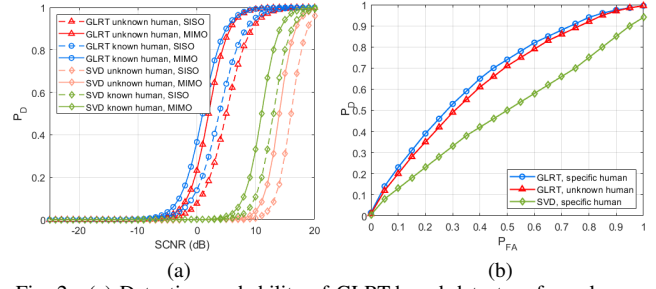


Fig. 2. (a) Detection probability of GLRT-based detectors for unknown and specific living person detection scenarios, and SVD method in [6] versus SCNR using SISO and 2×2 MIMO radar, $P_{FA} = 0.01$. (b) The ROC curves of the proposed detectors and SVD method using 2×2 MIMO radar, SCNR = -5 dB.

where $T_{\text{GLRT}}(\mathbf{y})$ the test statistic, $\mathbf{S}_1$ and $\mathbf{S}_0$ the delayed signal matrices given in (11), and γ_{GLRT} the decision threshold determined by the NP criterion.

Proof. The proof follows from (18) and is omitted. $\square$

The false alarm probability and the detection probability of the non-coherent GLRT-based MIMO radar living person detectors in (19) and (17) can be obtained by substituting the corresponding test statistic $T(\mathbf{y})$ and threshold γ into

$$P_{FA} = P(T(\mathbf{y}) > \gamma | H_0), \quad (20)$$

and

$$P_D = P(T(\mathbf{y}) > \gamma | H_1). \quad (21)$$

IV. NUMERICAL EXPERIMENTS

Although the theoretical results are derived under a general signal model, which does not depend on specific geometric configurations, due to space limitations, this section provides numerical examples to evaluate the proposed detectors for living person detection using MIMO radar. Consider a MIMO radar configuration with M transmitters and N receivers. In a two-dimensional Cartesian coordinate system, the antenna elements are placed uniformly along a circle of radius 10 m, centered at (0.1, 0.1). The angle of the m th transmitter is $\varphi_{tm} = 2\pi(m-1)/M + \varphi_{t0}$, $m = 1, 2, \dots, M$, and that of the n th receiver is $\varphi_{rn} = 2\pi(n-1)/N + \varphi_{r0}$, $n = 1, 2, \dots, N$, with $\varphi_{t0} = 0$ and $\varphi_{r0} = \pi/N$. Unless otherwise stated, this section considers a MIMO radar configuration with $M = 2$ transmitters and $N = 2$ receivers. The living person is assumed to be at (0.1, 0.1) m with reflection coefficient $\xi_{nm} = 0.1$, chest displacement $A_r = 0.01$ m, respiration frequency $f_r = 0.3$ Hz, and relative amplitude attenuation factor $\zeta_{nm} = 1$. The (17) and (19) are obtained by searching f_r over 0.1–2 Hz and A_r over 1–100 mm. The m th transmitted waveform is set as [18] $s_m(kT_s) = (2/T^2)^{1/4} e^{-\frac{\pi(kT_s)^2}{T^2}} e^{-j2\pi m f_\Delta kT_s}$, $0 < kT_s < T$, where pulse width $T = 5$ s, sampling frequency $f_s = 800$ Hz, and frequency offset $f_\Delta = 100$ Hz, ensuring approximate orthogonality among transmitted signals. Each transmitter radiates energy $E_m = E/M = 10^2/M$. The SCNR is defined as $\text{SCNR} = 10 \log \frac{\sum_{n=1}^N \sum_{m=1}^M |\xi|^2 E_m / d_{nm}^4}{N \sigma_w^2}$.

A. Performance Evaluation for Detectors

Fig. 2a shows the detection probability P_D in (21) versus SCNR for the proposed non-coherent GLRT-based detectors with known and unknown human reflection. The carrier

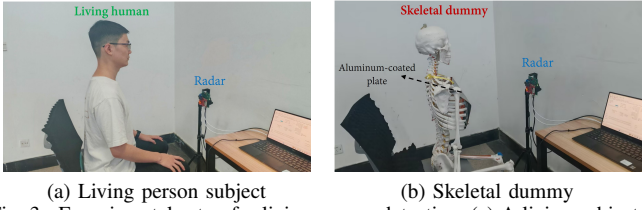


Fig. 3. Experimental setup for living person detection. (a) A living subject is tested under normal and shallow breathing scenarios. (b) A skeletal dummy is used as the non-living target, with a 20×20 cm aluminum-coated plate attached to the chest to block electromagnetic penetration.

frequency is $f_c = 1$ GHz with a false alarm probability $P_{FA} = 0.01$. In the SISO case, the transmitter and receiver are located at (10, 0) m and (0, 10) m, respectively. Solid lines denote MIMO radar results, while dashed lines represent SISO radar. Since this is the first work to propose two non-coherent MIMO radar-based detectors that differentiate a living person from a non-living object, the widely used SVD-based method in [6] is included for comparison. The results show that P_D increases with SCNR, and the proposed GLRT detectors consistently outperform the SVD method in all cases. The non-coherent GLRT-based detector with known human reflection shows better performance than the GLRT-based detector with unknown human reflection. Furthermore, MIMO radar consistently provides a significant performance gain over SISO radar, confirming the advantages of MIMO configurations for living person presence detection. Under the same MIMO radar setup as in Fig. 2a, Fig. 2b depicts the receiver operating characteristic (ROC) curves of the GLRT-based detectors with known and unknown human reflection at an SCNR of -5 dB. These results are consistent with those in Fig. 2a.

B. Experimental Example

In this section, we validate the effectiveness of the proposed living person presence detectors through experimental measurements, focusing on the more challenging scenario of an unknown living person. The study was approved by the Ethics Committee of the University of Electronic Science and Technology of China, Chengdu, Sichuan (IRB: 106142023081726465).

The experimental setup for living person detection is illustrated in Fig. 3, where (a) shows a living subject under normal and shallow breathing scenarios, and (b) shows a skeletal dummy. A TI AWR1642 mmWave radar is employed, transmitting frequency-modulated continuous wave (FMCW) signals over multiple chirps and frames. The radar configuration and signal parameters are summarized in Table I. To prevent electromagnetic penetration through the dummy, a 20×20 cm aluminum-coated plate is placed on its chest, and a microwave absorbing panel is positioned behind the target to suppress unwanted reflections.

Preprocessing is performed to calibrate the target position and ensure consistency with the experimental setup. The procedure includes three steps: (a) distance calibration, confirming both subjects are within the same range bin; (b) phase extraction, obtaining the phase information of the radar signal in the target range bin; and (c) baseline drift removal, applying

TABLE I
SYSTEM PARAMETER CONFIGURATION

Parameter	Value
Number of transmit antennas (M)	2
Number of receive antennas (N)	2
Start frequency (f_0)	77 GHz
Frequency slope	60.019 MHz/ μ s
Sampling rate (f_s)	10000 ksp/s
Number of frames	1000
Number of chirp loops	10/frame
ADC samples	256/chirp

a two-step cubic spline interpolation to eliminate drift and trend components.

Each volunteer performs 60 trials, including 20 trials for each of the two living-person scenarios and 20 trials for the skeletal dummy. Across three volunteers, a total of 180 experiments are conducted. Based on the hypothesis testing framework in (9), the radar measurements are analyzed using the GLRT-based detector described in Lemma 2 to distinguish between a living subject and a dummy. With the false alarm probability set to 0.1, the detection results for normal and shallow breathing are summarized in Table II. The GLRT-based detector achieves a detection probability of 0.8 under normal breathing, demonstrating its effectiveness in living person detection.

1) *Discussion and Future Work:* The detection performance of the proposed detector degrades under shallow breathing conditions. This is because normal respiration induces sufficiently distinct distributions under the two hypotheses, whereas shallow breathing leads to much closer distributions, which significantly challenges the GLRT-based detector. These results validate the effectiveness of the proposed detector for normal respiration while indicating that the development of more robust methods capable of handling weak respiration signals remains an important direction for future work.

TABLE II
GLRT-BASED DETECTOR AND SVD IN DISTINGUISHING A LIVING PERSON WITH NORMAL AND SHALLOW BREATHING FROM A DUMMY

state	GLRT	SVD
Sc.1 Normal versus Dummy	0.8	0.2
Sc.2 Shallow versus Dummy	0.1	0.1

V. CONCLUSION

This paper investigated living person presence sensing using MIMO radar system. A signal model for MIMO radar reception with distinct reflection coefficients across different propagation paths and hypotheses was developed, explicitly accounting for unknown respiration parameters, time-varying delays, and multi-dimensional reflections. Two non-coherent detectors for known and unknown human reflection scenarios were proposed for the first time, effectively addressing human reflection uncertainties. Theoretical analysis, numerical simulations, and experimental results validated the effectiveness of the proposed detectors and demonstrated the advantages of MIMO radar in living person detection, highlighting its potential for practical applications.

REFERENCES

- [1] L. Scalise, I. Ercoli, P. Marchionni, and E. P. Tomasini, "Measurement of respiration rate in preterm infants by laser Doppler vibrometry," in *2011 IEEE International Symposium on Medical Measurements and Applications*, 2011, pp. 657–661.
- [2] A. Qayyum, A. S. Malik, A. N. Shuaibu, and N. Nasir, "Estimation of non-contact smartphone video-based vital sign monitoring using filtering and standard color conversion techniques," in *2017 IEEE Life Sciences Conference (LSC)*, 2017, pp. 202–205.
- [3] M. Avraham, Y. Nemirovsky, T. Blank, G. Golan, and Y. Nemirovsky, "Toward an accurate IR remote sensing of body temperature radiometer based on a novel IR sensing system dubbed digital TMOS," *Micromachines*, vol. 13, p. 703, Apr. 2022.
- [4] J. Lin, "Noninvasive microwave measurement of respiration," *Proceedings of the IEEE*, vol. 63, no. 10, pp. 1530–1530, 1975.
- [5] C. Li, V. M. Lubecke, O. Boric-Lubecke, and J. Lin, "A review on recent advances in Doppler radar sensors for noncontact healthcare monitoring," *IEEE Transactions on Microwave Theory and Techniques*, vol. 61, no. 5, pp. 2046–2060, 2013.
- [6] B. P. A. Rohman, M. B. Andra, and M. Nishimoto, "Through-the-wall human respiration detection using UWB impulse radar on hovering drone," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 6572–6584, 2021.
- [7] P. Stockel, P. Wallrath, R. Herschel, and N. Pohl, "Detection and monitoring of people in collapsed buildings using a rotating radar on a UAV," *IEEE Transactions on Radar Systems*, vol. 2, pp. 13–23, 2024.
- [8] W. Han, S. Dai, and M. R. Yuce, "Real-time contactless respiration monitoring from a radar sensor using image processing method," *IEEE Sensors Journal*, vol. 22, no. 19, pp. 19 020–19 029, 2022.
- [9] P. Wang, A. Bian, B. Yu, and Q. He, "Detection of a Living Person With Unknown Reflection and Respiration Using MIMO Radar," *IEEE Transactions on Signal Processing*, vol. 73, pp. 2603–2615, 2025.
- [10] P. Wang, Q. He, and H. Li, "MIMO Radar Joint Heart Rate and Respiratory Rate Estimation and Performance Bound Analysis," *IEEE Signal Processing Letters*, vol. 32, pp. 1266–1270, 2025.
- [11] H. Alidoustaghdam, M. Chen, B. Willetts, K. Mao, A. Kokkeler, and Y. Miao, "Enhancing vital sign estimation performance of FMCW MIMO radar by prior human shape recognition," in *2023 IEEE International Conference on Communications Workshops (ICC Workshops)*, 2023, pp. 1707–1711.
- [12] S. Jiang, A. Alkhateeb, D. W. Bliss, and Y. Rong, "Vision-guided MIMO radar beamforming for enhanced vital signs detection in crowds," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 4, pp. 4640–4649, 2024.
- [13] M.-Z. Poh, D. J. McDuff, and R. W. Picard, "Non-contact, automated cardiac pulse measurements using video imaging and blind source separation," *Opt Express*, vol. 18, no. 10, pp. 10 762–10 774, May 2010.
- [14] D. K. Kim, K. W. Jang, and J.-H. Chang, "A new statistical voice activity detection based on UMP test," *IEEE Signal Processing Letters*, vol. 14, no. 11, pp. 891–894, Nov. 2007.
- [15] S. M. M. Kay, "Fundamentals of statistical signal processing, vol. II: Detection theory," *Detection Theory*, no. 4, 1993.
- [16] S. Kay and G. Boudreaux-Bartels, "On the optimality of the Wigner distribution for detection," in *ICASSP '85. IEEE International Conference on Acoustics, Speech, and Signal Processing*, vol. 10, 1985, pp. 1017–1020.
- [17] P. Wang and Q. He, "Quantized Quickest Change Detection with Unknown Parameters Using Cloud MIMO Radar," *IEEE Journal of Selected Topics in Signal Processing*, pp. 1–13, 2025.
- [18] Q. He, R. S. Blum, and A. M. Haimovich, "Noncoherent MIMO radar for location and velocity estimation: More antennas means better performance," *IEEE Transactions on Signal Processing*, vol. 58, no. 7, pp. 3661–3680, 2010.